Lie Groups — Journals

November 14, 2026



1 Journal 1

We will discuss several definitions of a smooth manifold on the special case that the manifold X is a euclidean manifold (embedded manifold), namely $X \subseteq \mathbb{R}^n$ for some $n \in \mathbb{N}$. Note that $\mathbb{R}^n$ is a T4 topological space of 2nd countability, therefore if A is an atlas on X then (X, A), or for the sake of convenience A alone, is a smooth manifold. When discussing embedded manifolds, there are several definitions, they are generally give substance to the concept of a manifold.

The first definition is Guillemin & Pollacks (GP) definition, as shown in their book "Differential Topology".

Definition 1.1 (GP smooth manifold)**.** Let $M \subseteq \mathbb{R}^n$ be a set and $k \in \mathbb{N}$. We say that M is a k-dimensional smooth manifold if that for any point $p \in M$ there is an open neighborhood $p \in U \subseteq \mathbb{R}^n$ and an open set $W \subseteq \mathbb{R}^k$ with function $F : U \to W$ diffeomorphism, meaning that its smooth and invertible. We define the function $f = F|_{M \cap U}$ as a smooth function from open subset of the manifold M.

Another version which we consider is defined by its parameterizations,

Definition 1.2 (Locally smooth parameterization manifold)**.** a set $M \subseteq \mathbb{R}^n$ together with a $k \in \mathbb{N}$ is called a smooth manifold if for any $p \in M$ there is an open set $U \subseteq M$ (open in the sub-topology of $\mathbb{R}^n$ induced by M), and an open set $W \subseteq \mathbb{R}^k$ such that there exists a function $\varphi : W \to U$ smooth (in the standard sense), open, invertible and with rank $D_x = k$ for any $x \in W$, namely a smooth local parameterization.

While both definitions may seem different, it can be shown using the inverse function theorem that they are equivalent.

Whitney embedding theorem states that any smooth manifold (X, A) can be embedded in $\mathbb{R}^{2k}$ space where $k = \dim(A, X)$, giving us motivation to compare our definition of a manifold in euclidean space to the definitions stated above.

Proposition 1.3. *If $M \subseteq \mathbb{R}^n$ is a GP k-dimensional smooth manifold and for any $p \in M$ let $f_p : U_p \to W_p$ be a diffeomorphism, where $U_p \subseteq M, W_p \subseteq \mathbb{R}^k$ open. Then $A = \{f_p \mid p \in M\}$ is an atlas, and in particular (M, A) is a smooth manifold.*

Proof. Clear by our definition of f_p as a restriction of a smooth function F_p and compositions. □

This result together with Whitney embedding theorem lead us to the conclusion that a manifold is a smooth manifold iff its a GP manifold (up to embedding).

2 Journal 2

We will discuss the claim that $\exp : L(G) \to G$ is a local homeomorphism, and in particular the fact that $\ker \exp$ is a discrete set. In general, the following theorem is known about embedded manifolds,

Theorem 2.1 (Preimage theorem). *If y is a regular value of $f : X \to Y$, then $f^{-1}(Y)$ is a submanifold of X with* $\dim f^{-1}(y) = \dim X - \dim Y$.

where,

Definition 2.2 (Regular value). If $f : X \to Y$ is a smooth map between manifolds, then $y \in Y$ is called a regular value if $df_x : T_x(X) \to T_y(Y)$ is surjective at every point $x \in f^{-1}(y)$.

If a function is locally homeomorphism, then in particular it is locally surjective, and defines a preimage manifold with $\dim f^{-1}|_U(y) = \dim X - \dim T_e X = 0$, namely it is a discrete set of points. While this is a very complex way to perceive this result, it binds the behavior of the exponential map to intersection-theory, and open the question about these two topics. Let us proceed to prove the claim in discussion directly.

Proposition 2.3. *If G has a Lie group structure, and* $\exp : L(M) \to M$ *is the exponential function, then* $\exp^{-1}(g)$ *is discrete for any $g \in G$.*

Proof. If $g \notin G_e$ then $\exp^{-1}(g) = \emptyset$, then let us assume that $\exp^{-1}(g)$ is non-empty, in particular $g \in G_e$. Assume for the sake of contradiction that a point $x \in \exp^{-1}(g)$ is non-discrete, namely there exist a sequence $\{x_n\} \subseteq \exp^{-1}(g)$ with $x_n \to x$. $\exp$ is a diffeomorphism, then let $\varphi : \overset{\subseteq G}{U} \to \overset{\subseteq L(G)}{W}$ be a chart, then $h = \varphi^{-1} \circ \exp$ is a diffeomorphism, but $\exp(x_n) = g$ therefore $h(x_n)$ is constant for any n, a contradiction. □

3 Journal 3

We will bring a full and formal proof of the closed subgroup theorem, assuming that the three lemmas Adams formulate hold.

Theorem 3.1 (Closed subgroup)**.** *If G is a lie group and $H \leq G$ is a closed subgroup, then H assumes a smooth manifold structure.*

Proof. Denote V the vector space for which the atlas maps G locally. By the third lemma there is a vector space $W \leq V$ such that $\exp W \subseteq H$ is a neighborhood of 0, in particular there is an open neighborhood of e in H, denoted by U. exp is a local diffeomorphism, by restricting exp to W or a smaller set Ω, we can assume that $\exp \Omega = U$ and $\exp|_\Omega : \Omega \to U$ is invertible. Note that to achieve $\Omega \subseteq W, U \subseteq \exp W$ we can use the inverse map theorem. Define $\varphi : U \to \Omega$ as the inverse of the restriction of exp, we get a function of a neighborhood of e in H. For any $p \in H$, define $\varphi_p : U_p \to \Omega$. Recall that the group multiplication is smooth, therefore the preimage of an open neighborhood of p, small enough, for the map $x \mapsto x \cdot p$, yields an open set such that $U_p = pU$ (where we assume that U is already small enough without loss of generality). Then $\varphi_p(x) = \varphi(p^{-1}x)$ is exactly composition of smooth maps therefore smooth, and is a chart around p. Construct $A = \{\varphi_p : U_p \to \Omega \mid p \in H\}$, a collection of smooth maps such that $\bigcup_{p \in H} \operatorname{dom} \varphi_p = H$, therefore (H, A) is indeed a smooth map. □